

UML Data Modelling

Carla Teixeira Lopes

Bases de Dados

Licenciatura em Engenharia Informática e Computação, FEUP+FCUP

Based on Jennifer Widom slides

Key concepts

Classes

Composition & Aggregation

Associations

Constraints

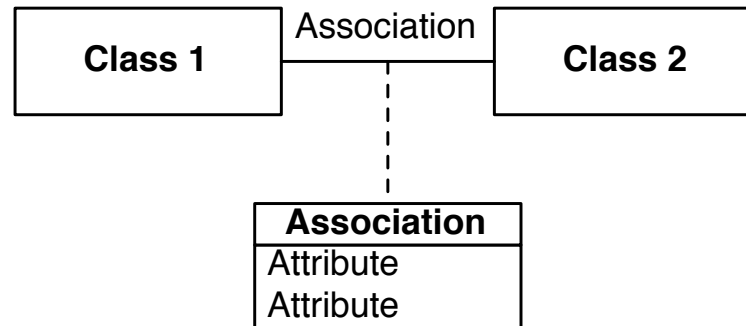
Association Classes

Derived Elements

Generalizations

Association Classes

Relationships between objects of two classes, *with attributes on relationships*

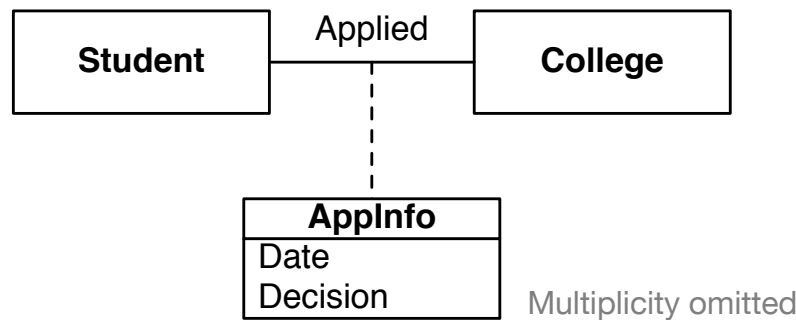


The name may be placed in the association, in the class or in both

It only captures **one** relationship between the two specific objects across the two classes

Association Classes

Suppose that, in the previous example, we also want to have the date the students applied to college and the decision.

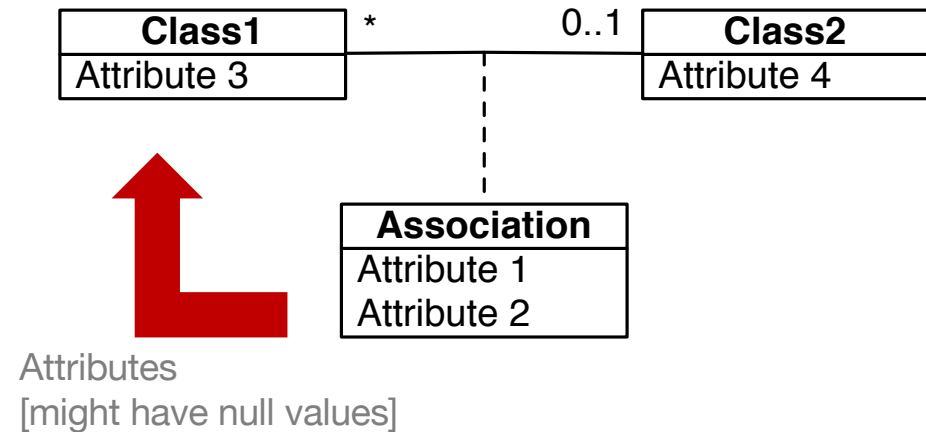
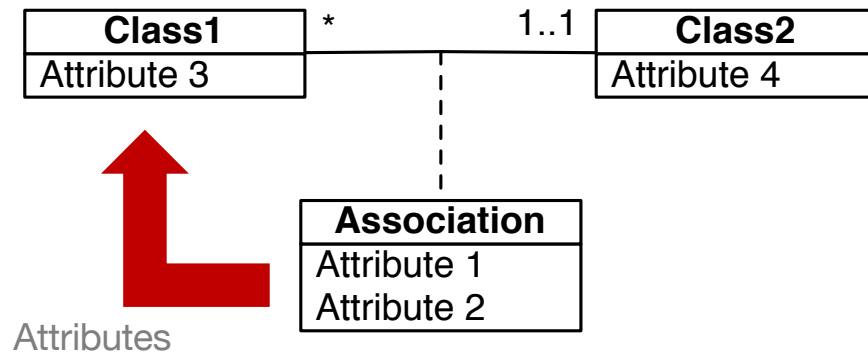


Doesn't allow students to apply multiple times to the same college.

How could we model this situation?

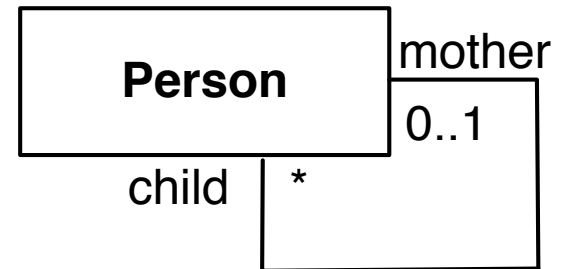
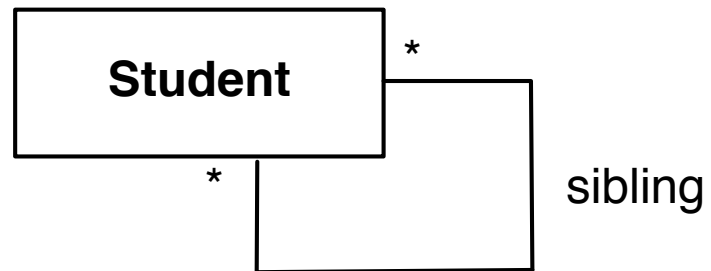
Eliminating Association Classes

Unnecessary if **0..1** or **1..1** multiplicity

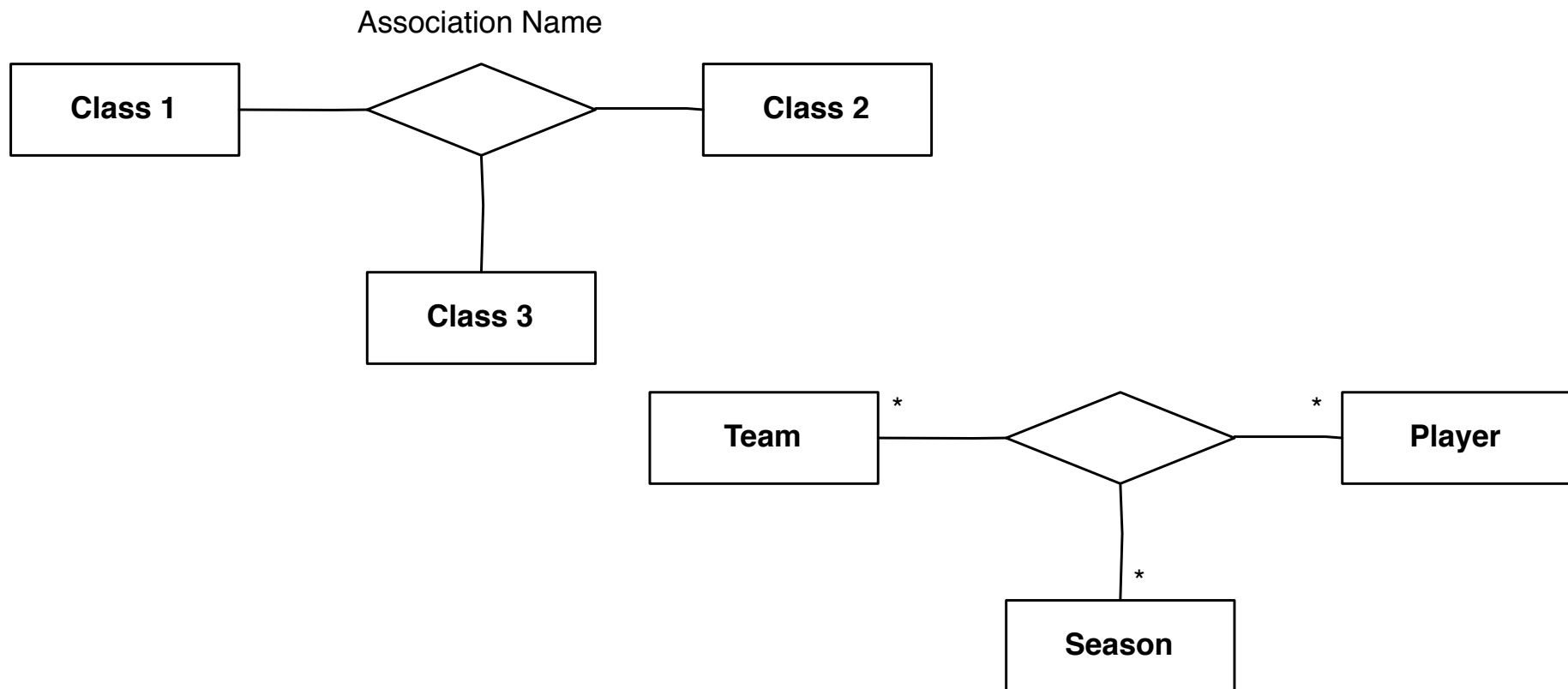


Self-Associations

Associations between a class and itself

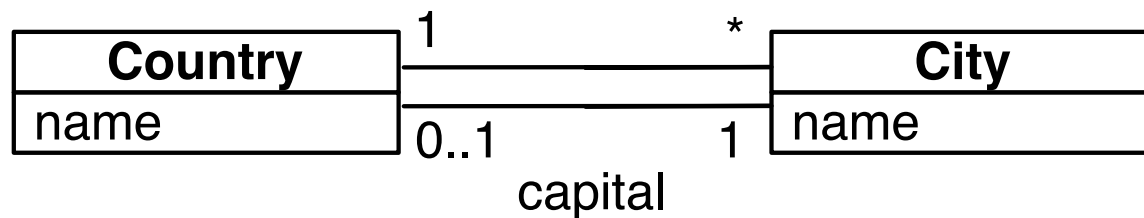
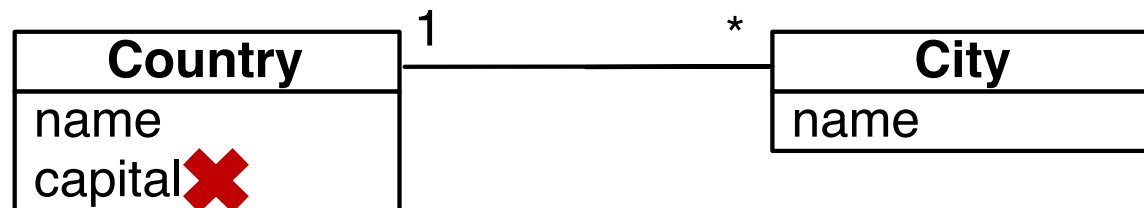


Associations n-arys

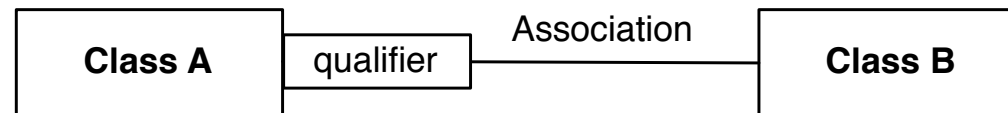


Association versus attribute

An attribute should never be a reference to a class



Qualified associations



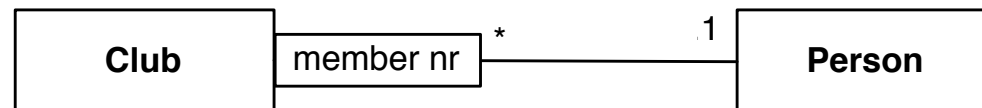
Qualifier

One or more attributes of an association used to navigate from A to B

“Access key” to B from an object of A

A person can be a member of several clubs

For each pair: Club + member nr



Exercise

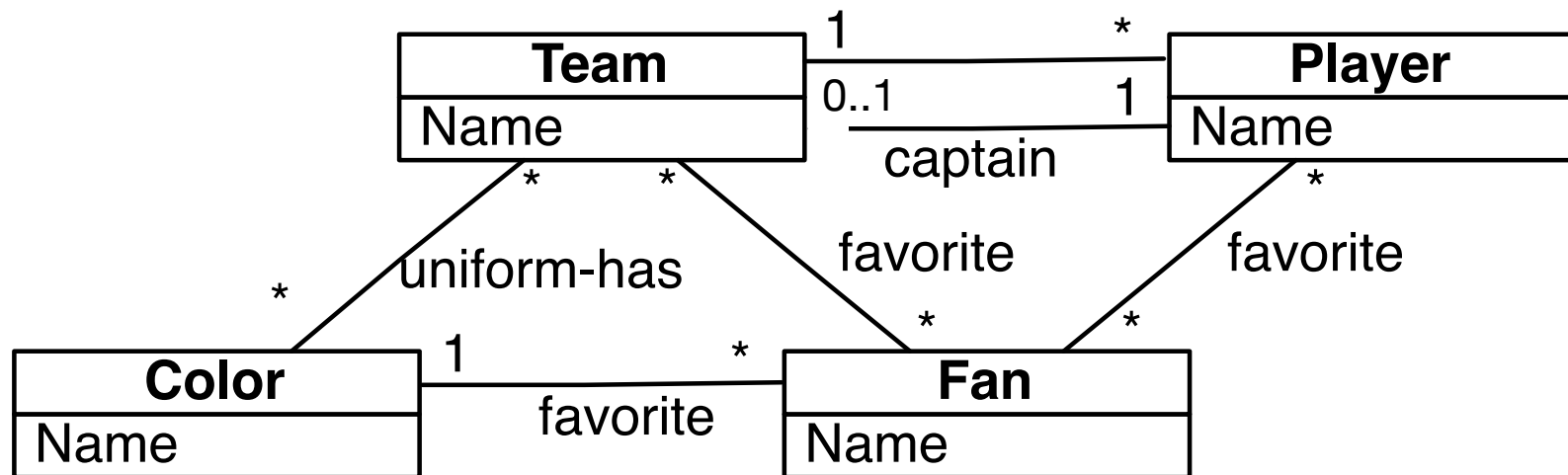
Draw a UML diagram for a database recording information about teams, players, and their fans, including:

For each team, its name, its players, its team captain (one of its players), and the colors of its uniform

For each player, his/her name

For each fan, his/her name, favorite teams, favorite players, and favorite color.

Exercise



Key concepts

Classes

Composition & Aggregation

Associations

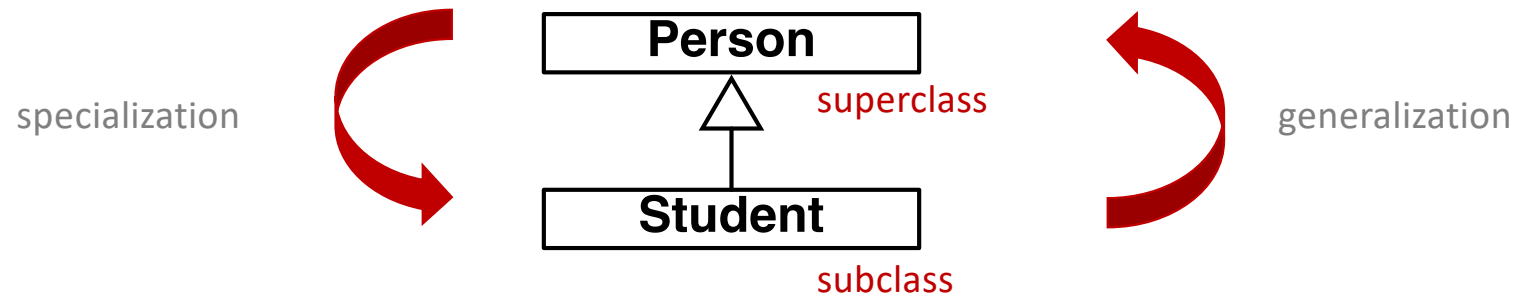
Constraints

Association Classes

Derived Elements

Generalizations

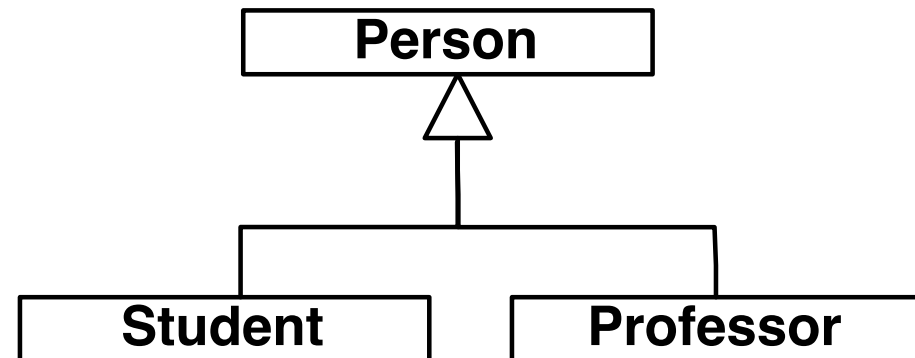
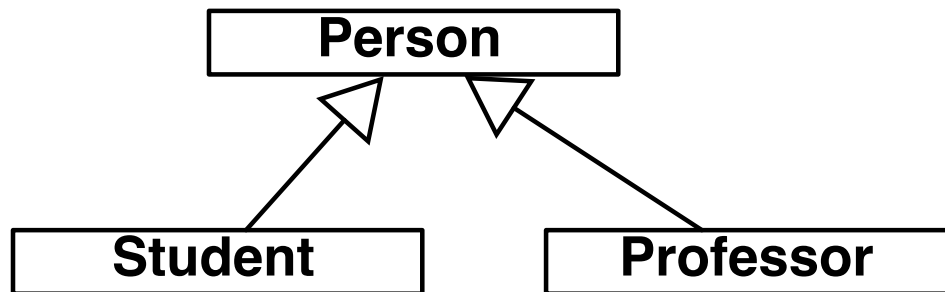
Generalizations



“is a” semantic relationship
A student “is a” person

The subclass inherits the properties (attributes and relationships) of the superclass and may add other

Generalizations – Alternative Notations



Generalizations Properties

Complete

If every object in the super class is in at least one subclass

Incomplete or partial

If it's not complete

Disjoint or Exclusive

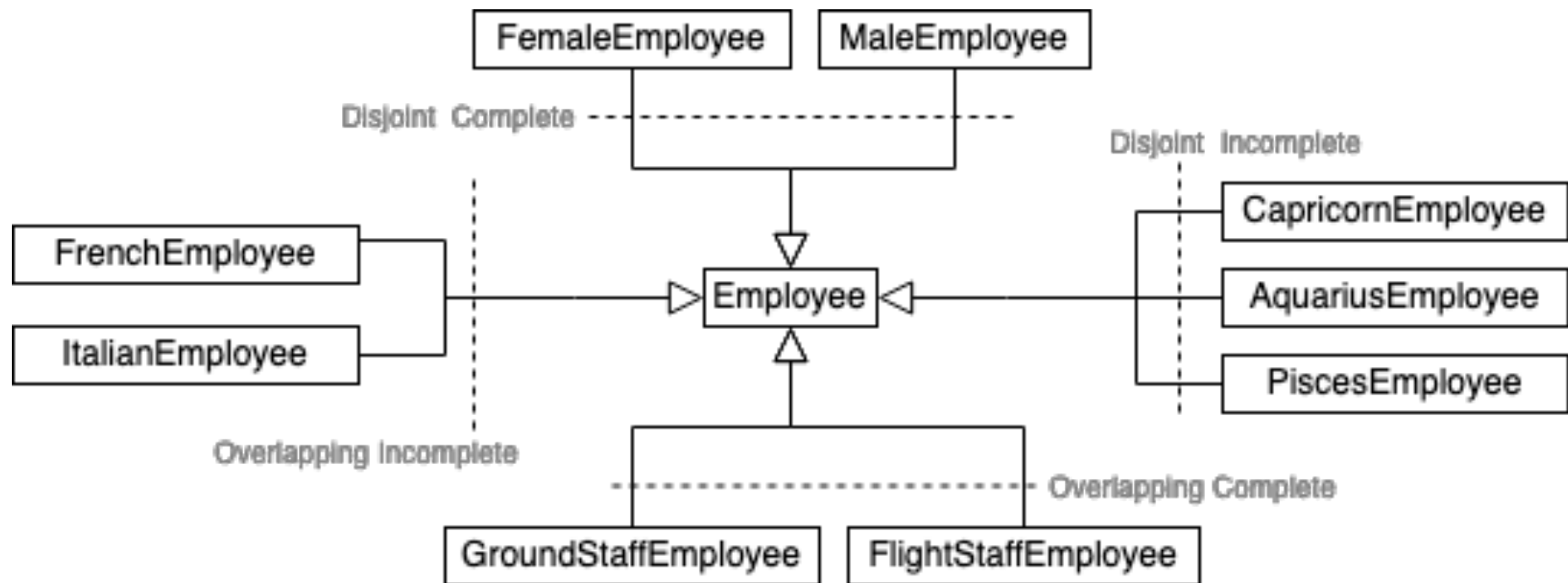
If every object is on at most one subclass

Overlapping

If it's not disjoint

We can have any
combination of the first
two with the second two.

Generalizations Properties



Martin Gogolla. UML and OCL in Conceptual Modeling. Conceptual Modelling – A Handbook. 2010

Key concepts

Classes

Composition & Aggregation

Associations

Constraints

Association Classes

Derived Elements

Generalizations

Aggregation

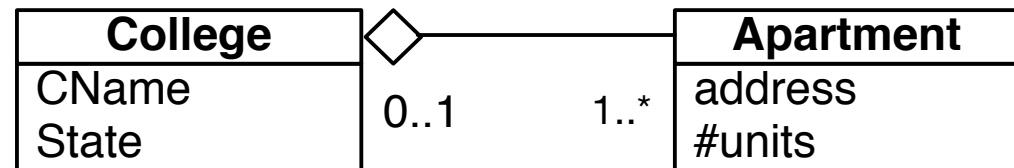
Special type of association

Models a “whole/part” relationship

Represents a “has-a” relationship

Does not link the lifetimes of the whole and its parts

0..1 is implicit



Aggregation

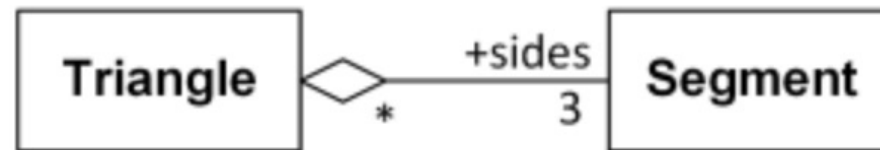
Binary association

Asymmetric

Only one end of association can be an aggregation

Shared part can be included in several composites

If some or all the composites are deleted, shared part may still exist



Triangle has 'sides' collection of three line Segments.
Each line Segment could be part of none, one, or several triangles.

Composition

Strongest form of aggregation

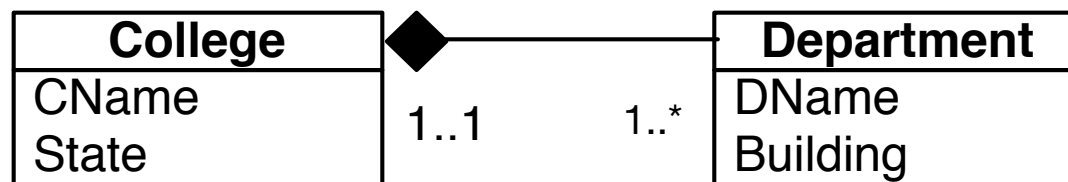
Strong ownership and coincident lifetime as part of the whole

An object may be a part of only one composite at a time

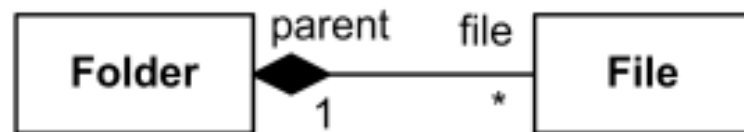
The whole is responsible for managing the creation and destruction of its parts

1..1 is implicit

Composition



College has 1 or more Departments and each Department belongs to exactly one College.
If College is closed, so are all of its Departments.



Folder can contain many files, while each File has exactly one Folder parent.
If Folder is deleted, all contained Files are deleted as well.

Key concepts

Classes

Composition & Aggregation

Associations

Constraints

Association Classes

Derived Elements

Generalizations

Constraints

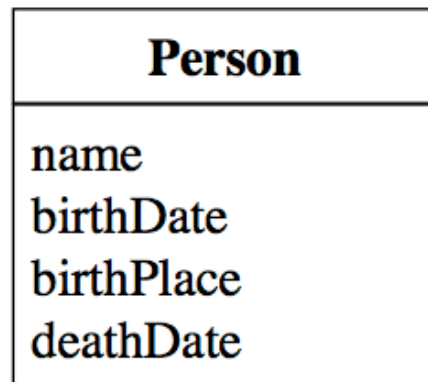
Specifies a condition that must be present in the system

It is indicated by

- an expression or text between brackets

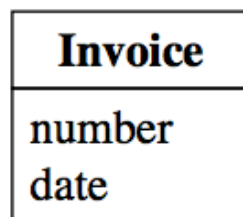
- note placed near (or connected by dotted lines to) the elements to which it relates

Constraints in classes



{candidate key: (name, birthDate, birthPlace)}

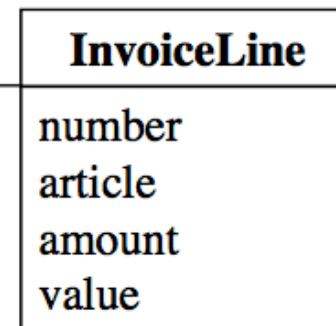
{deathDate > birthDate}



{candidate key: (number)}



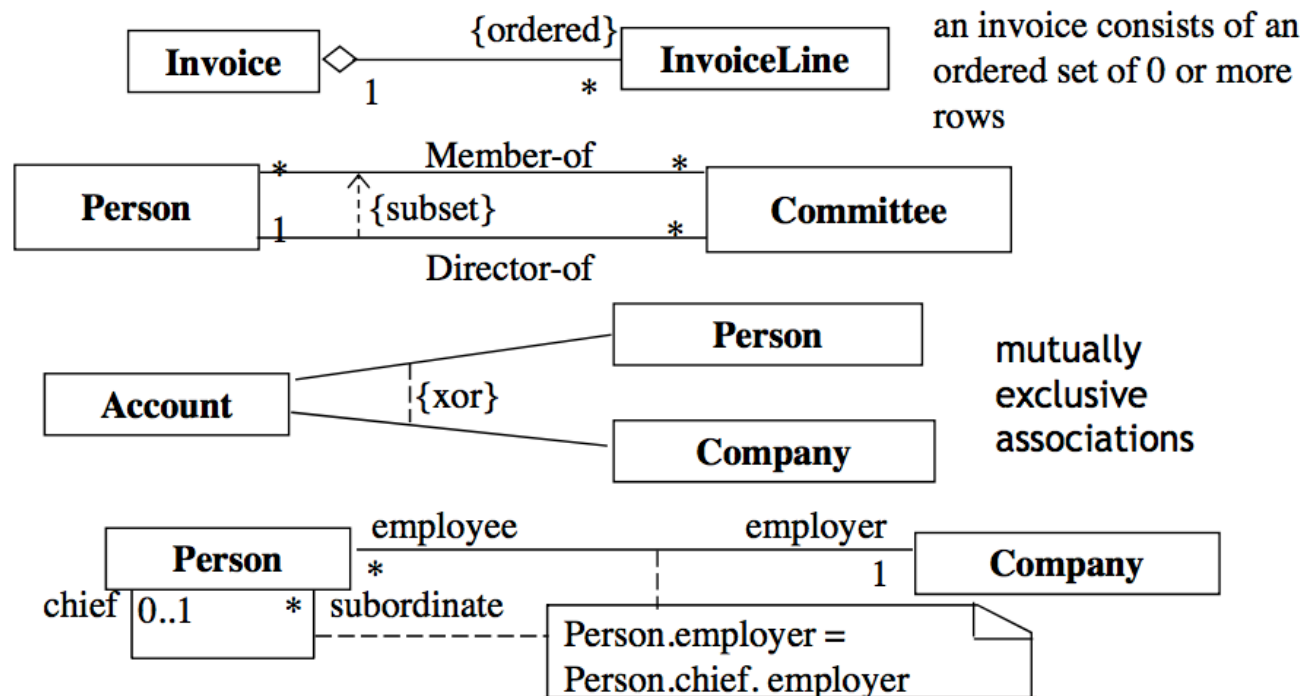
1



*

Credits: João Moreira

Constraints in associations



Credits: João Moreira

Key concepts

Classes

Composition & Aggregation

Associations

Constraints

Association Classes

Derived Elements

Generalizations

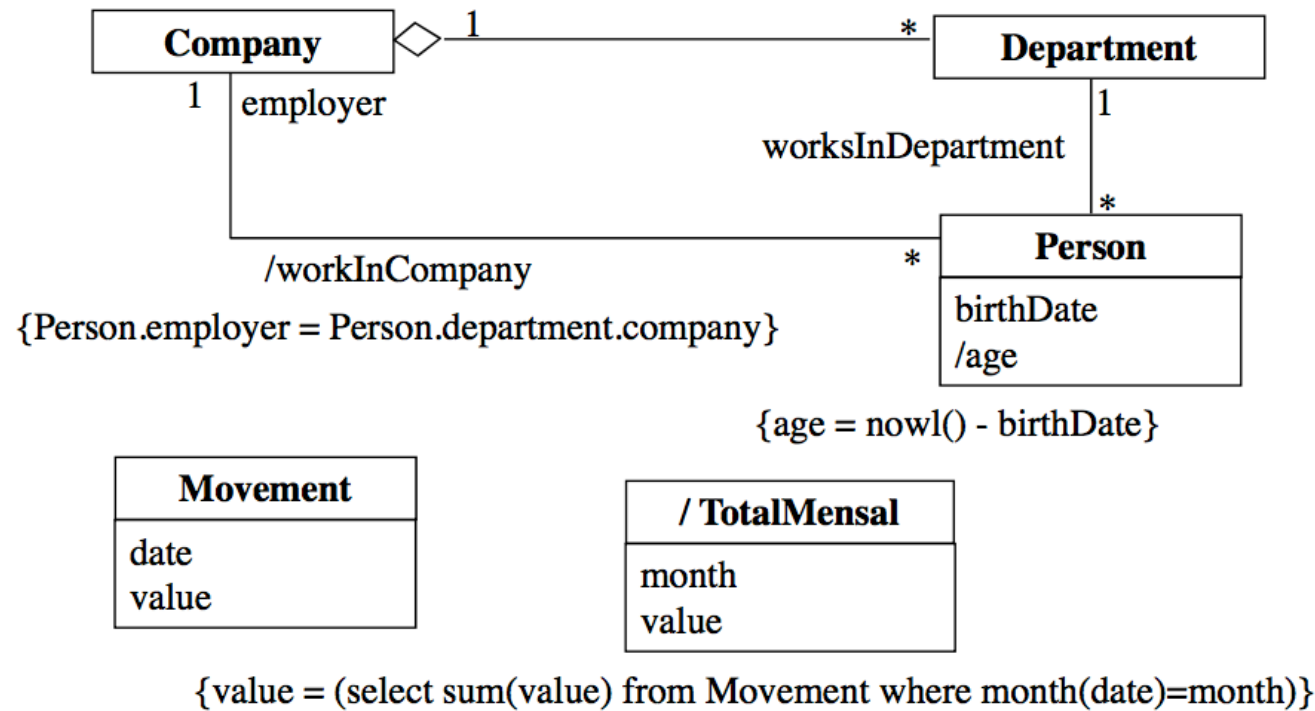
Derived Elements

Element (class, attribute or association) computed using other elements in the model

Notation: '/' before the name of the derived element

Usually have an associated constraint that relates them with other elements

Derived Elements



Higher-Level Database Design

Unified Modeling Language (UML)

Data modeling subset

Graphical

Key concepts

Classes

Associations

Association Classes

Generalizations

Composition & Aggregation

Constraints

Derived Elements

Can be translated to relations automatically

Wooclap time!

Any doubts?

Readings

Jeffrey Ullman, Jennifer Widom, A first course in Database Systems 3rd Edition
Section 4.7 - Unified Modeling Language